Ido Nitzan Hidekel, Ph.D. candidate

Ph.D. candidate in Deep Learning, Tel Aviv University • Lead, Deep Learning @ Dtect Vision

E-mail | Website | GitHub | Scholar | LinkedIn

Ph.D. student in Electrical Engineering specializing in **machine learning and deep learning**, with experience developing AI solutions from **problem formulation and research** through **data engineering, model development, evaluation, and deployment**. Lead Deep Learning Researcher at Dtect Vision and first author of papers accepted to **ICML 2026** and the **ICML 2026 Continual Learning (CATS) Workshop**.

Availability: 2-3 days per week

Experience

Lead, Deep Learning — Dtect Vision | Tel Aviv | 2025–present

Lead end-to-end algorithm development for large-scale synthetic-media detection. (Audio, Video and Imgae)

- Develop and evaluate deep-learning systems using vision foundation models, including architecture design, fine-tuning, calibration, and out-of-distribution evaluation.
- Built a **5M+ sample data pipeline** with leakage audits, source-aware sampling, and stratified evaluation.
- Implement distributed experimentation using **PyTorch DDP, GCP, Vertex AI, Docker, and W&B**.

Graduate Researcher — Geometric Deep Learning Lab, Tel Aviv University | 2024–present

Advisor: Prof. Dan Raviv

- Research optimization dynamics, generalization, spectral bias, continual adaptation, and foundation-model fine-tuning.
- Develop theoretical predictors and reproducible empirical pipelines across vision, language, and audio models.

Selected Publications

SONAR: Spectral-Contrastive Audio Residuals for Generalizable Deepfake Detection

Ido Nitzan Hidekel, Gal Lifshitz, Khen Cohen, Dan Raviv

ICML 2026, regular track. Paper | Code | Project

Developed an XLSR-based dual-path detector combining content and learnable residual representations. Achieved 1.45% EER on ASVspoof 2021 DF and 5.43% EER on In-the-Wild, with robustness to codec and bandwidth shifts.

Catastrophic Forgetting is Low-Rank: A Function-Space Theory for Continual Adaptation

Ido Nitzan Hidekel, Dan Raviv

ICML 2026 CATS Workshop — Continual Adaptation at Scale: Towards Sustainable AI. Paper

Developed a function-space predictor for catastrophic forgetting and showed that forgetting concentrates in a small number of dominant functional modes, motivating targeted regularization methods.

Education

Ph.D., Electrical Engineering (direct track) — Tel Aviv University | 2026–present

M.Sc., Electrical Engineering — Tel Aviv University | **GPA 95.5** | 2024–2026

B.Sc., Applied Mathematics — Tel Aviv University | **GPA 84** | 2020–2024

Technical Skills

Programming & ML. Python, PyTorch, Lightning, Claude Code, Lang Chain

Methods. Deep learning, foundation-model fine-tuning, representation learning, transfer learning, OOD detection, calibration, statistical evaluation.

Infrastructure. GCP, Vertex AI, PyTorch DDP, Docker, W&B, Linux, Git, distributed experimentation and large-scale data processing.